The Serial Physical Layer

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Abstract

This paper describes the RapidIO Serial Physical Layer (Part VI, 1x/4xLP-Serial). For those not familiar with the RapidIO interconnect technology, it is recommended that they first read *RapidIO: An Embedded Component System Network,* a white paper describing the open standard technology that can be found on the trade association's Web site (www.RapidIO.org).

Serial RapidIO extends the capability of the initial RapidIO interconnect technology by offering a fully serial, link protocol for those applications that require a memory-mapped protocol but with absolute minimal signal pins. Further, Serial RapidIO allows longer transmission distances using AC-coupled differential current mode signaling. This paper describes the high-level aspects of Serial RapidIO and some of its target applications, such as backplanes and processing farms.

Introduction

Sailboat designers have been faced with design tradeoffs in seeking the ultimate craft for many years. How does one design a boat to sail fast both into and away from the wind while withstanding often-treacherous sea conditions and still retain a comfortable passage? The idea is to maintain the most sail area possible while keeping boat weight to a minimum. Yet more sail area usually means longer masts. This results in the need for a larger keel to keep the boat righted. Larger keels mean greater boat weight. America's Cup boats have gotten around this problem by using bulbs positioned at the end of very long, thin keels to improve the righting moment. While they have achieved great speeds with these designs, the boats can only be used in "perfect" sailing conditions. In anything less, these boats suffer from broken masts, cracked keels and other major problems. Designers continue to deploy space-age materials and make dimensional adjustments, but in the end they still face design tradeoffs. The decisions from those tradeoffs have resulted in the variety of boats available today that are optimized for different uses, such as racing, lake cruising, or blue water circumnavigation, to name a few.

Tradeoffs are also the foundation of most engineering projects. One-size-fits-all does not work for many applications, and this is especially true for interconnect architectures.

Historical Perspective

When work on the RapidIO architecture began in 1997, it was quickly recognized that there were many tradeoffs to be made. The fundamental desire was to develop an interconnect architecture for processors and I/O devices which would preserve a simple memory mapped programming model while increasing the system level throughput and scalability. It was observed that with more available transistors, processor devices would incorporate more system-level integration. This would force the integration of both the processor mezzanine bus and I/O bus functionality into the same interconnect.

Work began on the RapidIO interconnect because traditional multi-drop busses had reached their maximum practical performance levels in both frequency and device drops. In order to reach higher levels of performance it was necessary to move away from busses to a point-to-point interconnect. Moving to a point-to-point interconnect meant that switches would be deployed to achieve multi-device connectivity. To keep the costs of switches in line, it was important to reduce the number of signal pins per link. More device-level system integration also meant more signal pins for various interfaces, further motivating a reduction in the number of signals dedicated to this interconnect.

An architectural tradeoff immediately surfaced: How narrow should the interconnect be? Both bandwidth and latency are major concerns for high performance microprocessors, so a narrower link interface would result in more clocks to transmit a transaction and thus longer latency. The resulting solution was to compromise on 8-bit or 16-bit widths (40 or 76 signals respectively). This was a good compromise for devices that were locally connected on a printed circuit board or for processor mezzanine card connectivity within a subsystem. However, for a point-to-point backplane there would be too many signals to practically communicate among several boards. Therefore, more than one physical interface would be needed.

For this reason the RapidIO specifications were partitioned to allow different physical mediums for interfacing devices while still largely preserving the associated transaction protocol. This would allow the device designer to preserve much of the controller designs and associated verification infrastructure while expanding the applications serviced by the interconnect protocol.

It was decided to first concentrate on the parallel RapidIO specification interface. At the same time, work was underway in the server arena to define a serial interface for system area networking (NGIO, FutureIO eventually leading to InfiniBand). For this reason it was decided to delay defining a narrower physical layer and see what fruit the InfiniBand effort would bear. As time moved on, several applications emerged where a lightweight memory mapped programming model was desired and the transaction set and overhead associated with either InfiniBand or Gigabit Ethernet made them less than satisfactory solutions.

Overview

The RapidIO 1x/4x LP-Serial (Link Protocol – Serial) was developed to address the need for a serial general-purpose memory mapped interconnect between devices within a system. Further, it allows ganging of four serial links for applications requiring higher link performance. While wider links are possible, the definition of this has been left for future work if warranted. The specification defines a full duplex serial physical layer interface (link) between devices using unidirectional differential signals in each direction. Serial RapidIO is useful for system applications where there is the need for low pin counts and not necessarily the highest performance achieved with wider parallel links like that found in the 8/16 LP-LVDS specification. The Serial RapidIO specification also defines a protocol for link management and packet transport over a link.

The RapidIO interconnect architecture is partitioned into a layered hierarchy of specifications (Figure 1) which include the Logical, Common Transport, and Physical layers. The Logical layer specifications define the operations and associated transactions by which end point processing elements communicate with each other. The Common Transport layer defines how transactions are routed from one end point processing element to another through switch processing elements. The Physical Layer defines how adjacent processing elements electrically connect to each other. Packets are formed through the combination of bit fields defined in the Logical, Common Transport, and Physical Layer specifications.

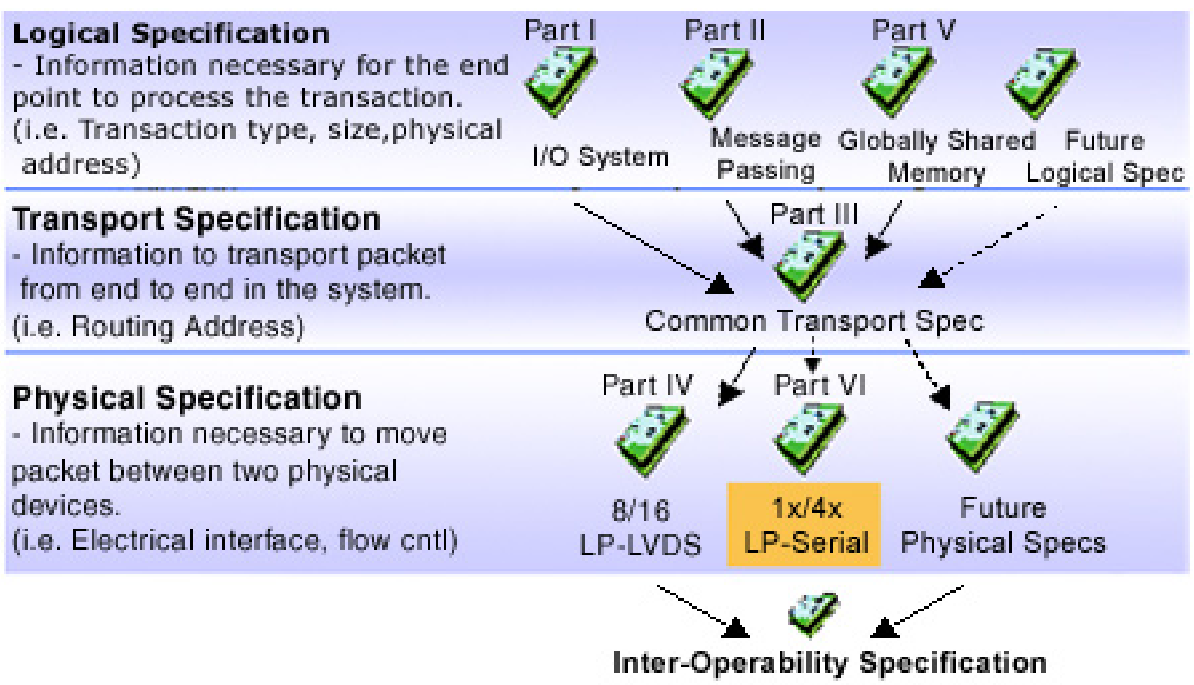

Figure 1:The RapidIO 1x/4x LP-Serial specification is a physical layer specification leveraging the same upper layer protocol as the 8/16 LP-LVDS parallel physical layer.

The RapidIO Physical Layer 1x/4x LP-Serial specification defines a protocol for packet delivery between serial RapidIO devices including packet and control symbol transmission, flow control, error management, and other device-to-device functions.

The 1x/4x LP-Serial physical layer specification has the following properties:

- Embeds the transmission clock with data using an 8B/10B-encoding scheme.
- Supports one serial differential pair, referred to as one lane, or four ganged serial differential pairs, referred to as four lanes, in each direction.
- Allows switching packets between RapidIO 1x/4x LP-Serial Ports and RapidIO Physical Layer 8/16 LP-LVDS ports without requiring packet manipulation.
- Employs similar retry and error recovery protocols as the RapidIO Physical Layer 8/16 LP-LVDS specification.
- Supports transmission rates of 1.25, 2.5, and 3.125 Gbaud (data rates of 1.0, 2.0, and 2.5 Gbps) per lane.

Applications

Serial RapidIO technology is useful in applications where a simple memory mapped load-store access model is desired. The advantage of the load-store model includes much lower transaction overhead and lower resulting communication latency between processing elements as compared to message based interconnects like InfiniBand and Gigabit Ethernet.

Applications for Serial RapidIO interconnect include:

- System control plane interface
  - Passing of DMA descriptors
  - Channel management
  - System management
  - Parameter passing
- Processor interface (DSP or Control processor)
  - Replacement of host port with narrow, lower pin interface
  - Connection point for DSP farms
  - Memory access intense applications
  - Interface to common memory controller for code/parameter passing
- Pin Constrained Applications
  - Backplanes

Figure 2 is an example of a generic system where both parallel and serial RapidIO interconnects are deployed. For this example parallel links are used within the processing subsystem to allow very high–performance, low-latency communications between processors and associated ASICs. The subsystems may be linked together across a backplane using serial RapidIO interconnect. The serial link reduces the number of signal traces and connectors that must be managed on the backplane. Connectivity is done through fabric cards. Redundant connections are shown in this example to eliminate a single point of failure. Some applications, like that of an IP-telephony DSP farm, are well served by the serial performance. A serial interconnect allows the high switch fan out required to connect the many DSP devices associated with this application.

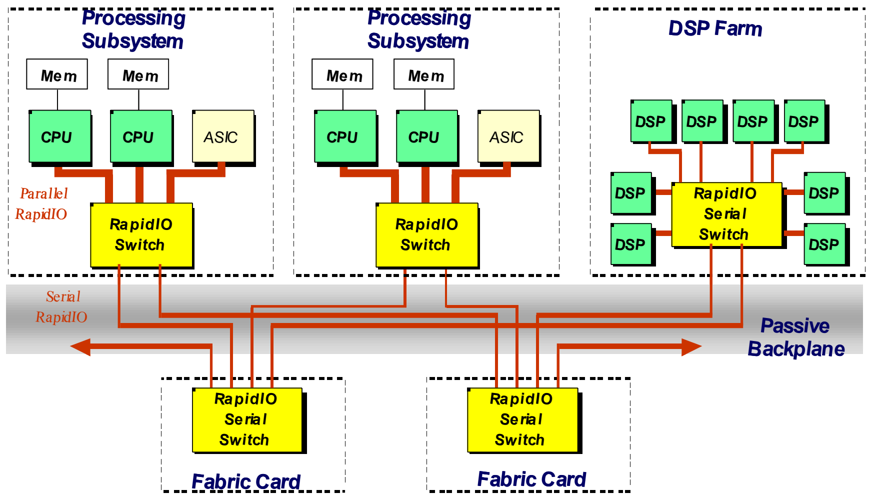

Figure 2: A typical system may deploy a combination of parallel and serial RapidIO links to achieve desired connectivity and performance.

System Topologies and Peer-to-Peer Communication

An advantage of RapidIO technology over other interconnects is the ability to support many different connection topologies. This includes simple daisy chains, spanning trees, and complex meshes. Unlike PCI and PCI derivatives, the RapidIO interconnect uses source-based addressing. Source-based addressing means that the packet contains fields for both the destination address and source address of the transaction. The source issues a transaction to an explicit destination. Switches use this destination route address to steer the packet through a system to the final destination. The destination uses the source address from the packet to respond to the packet if necessary.

Configuration of the system is done similar to that of PCI; however, additional capabilities are provided to allow multiple hosts to explore the system. Adjusting only the routing tables in the RapidIO switches can quickly modify the path through which transactions traverse the system. This can be useful for software to manage traffic flows and for redundant fail-over paths.

Ethernet in the Backplane

Because of the ubiquitous nature of Ethernet, it has found its way into the backplane of modern embedded systems. In such applications it is typically used as a message channel between boards or subsystems. Some advantages of using Ethernet include: readily available hardware, low pin count, existing software stacks and the associated software based error management. However, in many applications, the overhead associated with sending Ethernet transactions makes it a less than satisfactory choice because ethernet requires processors at each end to manage the protocol stack. For applications that send many small data payloads, the overhead associated with Ethernet packets becomes burdensome. It can also be difficult to guarantee deterministic throughput behavior. Lastly, Ethernet does not match up directly to the load-store behavior of processors.

For these applications, Serial RapidIO technology is better suited to the task allowing memory mapped, direct load-store communications without the need for software managed protocol stacks. This leaves the processors free to do more interesting tasks. The Serial RapidIO physical interface is compatible with commodity backplane technology. The transaction overhead is very low. Transaction transport is managed completely in hardware and error detection and the link protocol is robust.

Serial RapidIO Details

The Serial RapidIO Controller

Since the Serial RapidIO specification is only defined in the physical layer (the RapidIO technology defines the physical layer as the electrical interface and device to device link protocol), most of the controller remains the same. As a result, much of the design knowledge and verification infrastructure are preserved and eases system level switching between parallel and serial links.

During the initial development stages of the Serial RapidIO specifications it was decided to try and preserve as many of the concepts found in the RapidIO parallel specification as feasible. The parallel specification includes the concept of packets and in-band control symbols. These were delineated and differentiated by both a separate frame signal and an “S” bit in the header. In moving to a serial link it was decided to encode this delineation into spare characters (“K-codes”) found in the 8B/10B encoding. In this way, the sending device indicates to the receiving link partner the start of a packet, end of packet or embedded control symbol using these codes.

Link Protocol

The link level protocol used in the Serial RapidIO specification is similar to that found in the Part IV RapidIO 8/16 LP-LVDS parallel specification. A unique feature of parallel RapidIO technology is that packet transmission is managed on a link-by-link basis. In the past, with synchronous busses, a mastering device had explicit handshake signals with the target device. These signals indicated whether a transaction was acknowledged and accepted by the target device.

With a source synchronous interface like the RapidIO specification defines, it is not practical to rely on a synchronous handshake since the receive port of a link is decoupled from the sending port. Therefore, many interconnects have ignored this issue and rely on an end-to-end handshake to guarantee delivery. However, this has the disadvantage of not allowing precise detection and recovery of errors and forces far longer feedback loops for flow control.

To address this issue, the parallel RapidIO technology, and now Serial RapidIO technology, use embedded Control Symbols for link level communication between devices. Packets are explicitly tagged between each link with a sequence number otherwise known as AckID. The AckID is independent of the end-to-end transaction ID. Using Control Symbols, the receiving device indicates for each packet whether it has been received along with additional buffer status information. A receiving device can immediately detect a lost packet, and through Control Symbols, can re-synchronize with the sender and recover it without software intervention. The receiving device then forwards the packet to the next switch in the fabric, and so on, until the final target of the packet is reached.

Flow Control

Serial RapidIO technology allows for the possibility of longer transmission distances and thus longer loop latencies in providing feedback between the receiver and transmitter on a link. Therefore Serial increases the number of AckID values from 8 to 32. Additionally, the Serial RapidIO specification now defines a Transmitter-Controlled flow control scheme whereby the receiving port provides information to its link partner about the amount of buffer space it has available. With this information, the sending port can allocate the use of the receiving port's receive buffers based on the number and priority of packets that the sending port has waiting for transmission. The sending port does not have to be concerned that one or more of the packets shall be forced to retry.

PCS and PMA Layers

The Serial RapidIO specification uses a Physical Coding Sublayer (PCS) and Physical Media Attachment (PMA) sublayer to organize packets into a serial bit stream at the sending side and extract the bit stream at the receiving side (terminology adopted from IEEE 802.3).

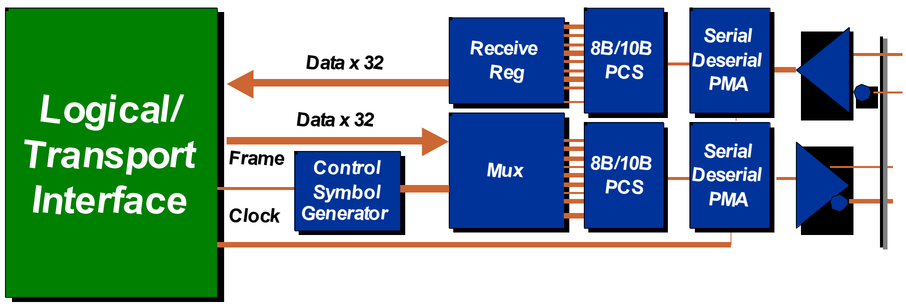

Figure 3: The RapidIO Serializer/Deserializer function integrates directly with the logic and transport layers

Besides encoding for transmission and decoding, the PCS function is also responsible for idle sequence generation, lane striping, lane alignment, and de-striping on reception. The PCS uses an 8B/10B encoding for transmission over the link.

The PCS layer also provides the mechanisms for automatically determining the operational mode of the port as either 1-lane or 4-lane, and provides for clock difference tolerance between the sender and receiver without requiring flow control.

The PMA function is responsible for serializing 10-bit parallel code-groups to/from a serial bit stream on a lane-by-lane basis. Upon receiving data, the PMA function provides alignment of the received bit stream to 10-bit code-group boundaries, independently on a lane-by-lane basis. It then provides a continuous stream of 10-bit code-groups to the PCS, one stream for each lane. The 10-bit code groups are not observable by layers higher than the PCS.

Electrical Interface

The RapidIO Technical Working Group decided that it was important to leverage standard signaling technology to reduce the barriers of usage. Serial RapidIO technology uses differential current steering drivers similar to that found in 802.3 XAUI, 10 gigabit Ethernet, Fibre Channel, and InfiniBand. This signaling technology was developed to drive long distances over cables or backplanes. To overcome the signal attenuation in such applications a large voltage swing is used. This comes at the expense of additional power consumption.

For the Serial RapidIO technology, two transmitter specifications were designated: a short run transmitter and a long run transmitter. The short run transmitter is used mainly for chip-to-chip connections either on the same printed circuit board or across a single connector such as that for a mezzanine card. The minimum swings of the short run specification reduce the overall power used by the transceivers. A user can further reduce the power by lowering the termination voltages.

The long run transmitter uses larger voltage swings that are capable of driving across backplanes. This allows a user to drive signals across two connectors and common printed circuit board material.

To ensure interoperability between drivers and receivers of different vendors and technologies, AC coupling must be used at the receiver input.

Maintenance and Error Management

The RapidIO interconnect is designed and targeted for high-performance, high-availability embedded infrastructure applications. These applications require robust and dependable operation from all aspects of the system, including the interconnect. For this reason, great attention has been paid in the RapidIO specification to the management of operational errors. This includes everything from bit errors, lost packets, out of sequence packets, and insane devices. Many interconnects rely on software protocol stacks for transaction and error management. Because the RapidIO technology is intended for very tightly coupled, low overhead, load/store applications, all of the detection mechanisms must be built into hardware. Further, most embedded applications rely on real-time behavior and do not have time to fall back to software for error recovery. Therefore, RapidIO has built-in hardware-based error recovery mechanisms to handle all but the most severe conditions. For severe failures, there are a variety of software notification mechanisms.

The Serial RapidIO specification maintains all of the error management mechanisms found in the parallel RapidIO specifications.

Summary

The 1x/4x LP-Serial specification extends the capabilities of the RapidIO specification suite to enable more applications with a common interconnect protocol. In choosing a RapidIO physical layer the following points should be kept in mind:

- 8/16 LP-LVDS - Parallel
    - Highest performance
    - Latency sensitive applications
    - High bandwidth
    - Short transmission distance
    - Higher pin count
    - Examples: High performance processor complex, Network processor complex, High performance backplane
- 1x/4x LP-Serial
    - Latency tolerant applications
    - Longer distance
    - Power and/or pin count sensitive applications
    - Examples: Backplane control plane/message channel, DSP farms, Chassis expansion

Serial RapidIO technology provides robust, industry-known signaling technology allowing longer distance transmission over common mediums. It preserves all of the error detection and recovery capabilities found in the parallel specification. The combination of the parallel interface and now serial interface offers more choices for the system designer to better address the tradeoffs found in a high performance embedded system.

Note: The parallel RapidIO specification, more information on the open-standard interconnect and a description of the trade association are available at www.RapidIO.org.